Machine Learning Approaches for Water Quality Index Prediction: A Systematic Review

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GitHub Repository: https://github.com/mk19409-prog/Literature-Review-on-WQI-ML
Detailed Literature Review Sheet: https://docs.google.com/spreadsheets/d/1uNeQQFx8vY5-hyzwO9BjR6l4s6zKMueg/edit?usp=sharing&ouid=106348839836730303077&rtpof=true&sd=true

I. Introduction

Water is an essential natural resource that supports ecological balance, sustains human livelihoods, and drives agricultural, industrial, and environmental systems. Surface water is one of the numerous water sources that can be utilized to meet the freshwater needs of all societies in the world, such as rivers, lakes and reservoirs [1]. Unregulated urbanization, industrial effluents, agricultural activities, and climate-induced events have contributed to the accumulation of major pollutants in waterways. Traditional ways of assessing water quality are often based on lab analysis and manual interpretation; these methods are precise but resource-intensive, time-consuming, and ineffective for real-time or large-scale monitoring. Water Quality Index (WQI) is a quantitative method used to evaluate the overall condition of a water body based on selected physical, chemical, and biological parameters [2].

Machine learning improves water quality assessment by modeling complex and nonlinear relationships that can adapt in different geographic, seasonal, and hydrological conditions [3]. Different ML models (such as Decision Trees, Random Forests, Support Vector Machines, Regression, and Gradient Boosting) and DL models (Artificial Neural Networks, Convolutional Neural Networks, Recurrent Neural Networks, Long Short-Term Memory networks, and transformers) are widely used for the prediction and forecasting of the water quality index [4], [5]. In addition, multi-task learning frameworks allow the simultaneous prediction of multiple WQIs for different water uses by sharing common feature representations while generating task-specific outputs, improving scalability and generalization across diverse contexts [6].

Most existing WQI models are region-specific (e.g., designed for Canada or Ireland) or source-specific (e.g., rivers or wetlands) and produce a single index based on fixed parameter weightings and thresholds. Such static designs lack flexibility for multi-context environments where water bodies simultaneously support diverse uses such as drinking, agriculture, fisheries etc. These have created a critical research gap for the further review of a universal and adaptive AI-intensive WQI model that will enable mass application to a wide range of purposes with modern sustainability demands.

II. Methodology

The selection of appropriate articles for this research followed a comprehensive and organized methodology to guarantee the integration of high-quality and relevant literature relating to Water Quality Index (WQI) prediction and classification through Machine Learning technique.

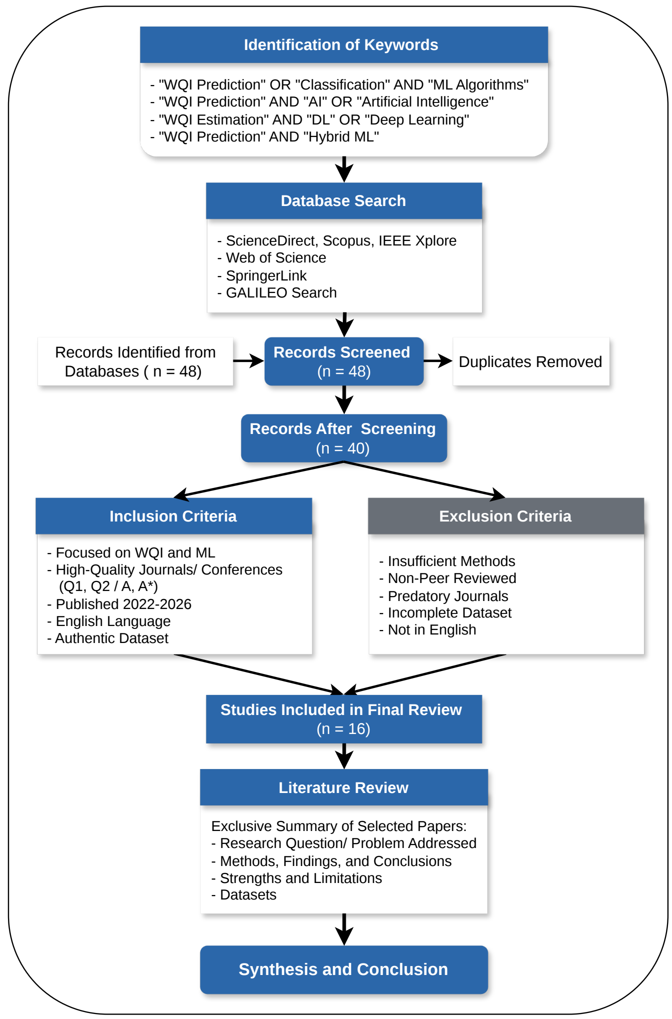


Fig. 1: Overall Pipeline of the Study Design

The whole process has multiple steps, such as finding keywords, accessing databases, initial screening, and using inclusion and exclusion criteria [2], as depicted in Figure 1.

(1) **Keyword Used** To find the relevant research papers, the first procedure is to define the appropriate keyword

searches. The following combinations of keywords have been used:

- Water Quality Index (WQI) Prediction/Classification and Machine Learning (ML) Algorithms
- Water Quality Index (WQI) prediction and Artificial Intelligence (AI)
- Water Quality Index (WQI) estimation and Deep Learning (DL)
- Water Quality Index (WQI) prediction and Hybrid Machine Learing Algorithms

(2) **Databases Accessed:** We used several well-known scholarly databases to find reputable, research-oriented publications. These types of databases publish peer-reviewed and trustworthy scientific research. The databases used for the current study include the following:

- ScienceDirect
- Scopus
- IEEE Xplore
- Web of Science
- SpringerLink
- Google Scholar
- GALILEO Search

Applying more than one database helps make sure that all relevant research in the domain of water quality prediction using machine learning is considered.

(3) **Initial Paper Identification**
An initial search was done based on the keywords and databases. This stage resulted in the finding of 48 research publications that seemed to be relevant based on their titles, abstracts, and keywords.

(4) **Screening of Relevant Research Papers**
The research papers that have been identified were subsequently reviewed to see if they were relevant to the research topic. At this point, duplicate research was taken out, and publications that didn't have anything to do with WQI prediction or categorization were also taken out. After this initial assessment, 40 papers were chosen for more review.

(5) **Inclusion criteria**
The following inclusion criteria were used to make sure that the research paper is relevant and high quality.
We have presented some inclusion criteria:

- Focused on Water Quality Index (WQI) prediction and classification using solor or hybrid machine learning.
- High-quality peer-reviewed journal (Elsevier, IEEE, springer (Minumum quartiles: Q1 and Q2)) and conference (Ranked: A and A (star)).
- Research Paper Published in Recent years (e.g, 2022, 2023, 2024, 2025 and 2026).
- Published in the English language.
- Recent/update methodological details.
- Dataset: Authentic sources, complete parameters for WQI.
- Initial selection Research papers: 48, selected based on our research (WQI): 40, finally selected: 16 journals (WQI related).

(6) **Exclusion Criteria**
Along with inclusion criteria, some exclusion rules were used to get rid of studies that weren't relevant or of low quality. The following are the exclusion criteria:

- Insufficient methodological details.
- Journal papers: not peer reviewed; conference papers: not list in the ranking.
- Predatory journals.
- Dataset: unauthenticated source, insufficient, and incomplete parameters.
- Not written in English.

(7) **Final Paper Selection**
After using the inclusion and exclusion criteria, mainly the most relevant and high-quality studies were kept. Out of the first 48 papers, 40 were chosen for further review, and then 16 high-quality research papers that were closely relevant to WQI prediction and classification were chosen for this study's extensive analysis.

III. Literature Review

Recent research on Water Quality Index (WQI) prediction shows a strong transition from conventional weighted arithmetic models toward machine learning (ML), deep learning (DL), and hybrid ensemble approaches [13]. Across the reviewed papers, the central problem is the limitation of traditional WQI frameworks, including rigid parameter weighting, inconsistent classification schemes, region-specific formulations, and poor generalization across environmental contexts [5], [14]. An exclusive summary of the literature review with most crucial seven papers are presented in Table I. A detailed literature review can be found from the access link to the Google Sheet.

Most studies relied on region-specific datasets, including:

- Kaggle Water Potability dataset (3276 samples)
- Telangana groundwater dataset (Kaggle)
- Coastal water data from catchments.ie (EPA, Ireland)

Input features typically included physicochemical water quality parameters such as pH, DO, BOD, COD, EC, TDS, turbidity, sulfate, hardness, salinity, and sometimes climate indicators. However, except catchment dataset, many datasets were small and geographically limited [11], [15].

Most studies applied supervised ML algorithms such as Random Forest (RF), Support Vector Machine (SVM), XGBoost, KNN, Decision Tree, and Artificial Neural Networks (ANN). Results comparison is summarized in the *Key findings and contributions* column of Table I. More recent works incorporated:

- Hybrid models (e.g., ANN + XGBoost with SHAP-based initialization – Bataineh et al., 2026) [7]
- Stacking ensemble models (Mohseni et al., 2024) [9]
- Advanced deep learning architectures (LSTM, optimized DL frameworks) (Han et al., 2025; Satish et al., 2024) [10]

TABLE I: Exclusive Summary of Selected Papers

Reference	Problem addressed	Methods	Findings and contributions	Limitations/Gaps	Dataset
(Bataineh et al., 2026) [7], [Q1, IEEE]	Comparison to traditional methods, a hybrid model incorporated XGBoost boosting with SHAP-based feature-informed neural network initialization can increase the accuracy of Water Quality Index prediction.	Models: ANN, XGBoost, and Hybrid ANN + XGBoost model (proposed model) Baseline models for comparison: Random Forest, Support Vector Regression, Standard ANN, Standalone XGBoost	When compared to random initialization, SHAP-based weight initialization enhanced neural network convergence and performance. Accuracy: 86.9%, F1-score: 84.9%, ROC-AUC: 89.4%	The dataset employed binary classification, although multiple classification is necessary in real life. There is no specific reference of potability measures.	Public Water Potability dataset from Kaggle:
(Nishat et al., 2025) [8], [Q1, Springer]	Dhaka rivers are severely contaminated by sewage, industrial discharge, and urban runoff; precise WQI forecasting is required for environmental management.	Machine Learning Models (14): ANN, RFR, Decision Tree Regression, Linear Regression, Ridge Regression, SGD Regressor, XGBoost and etc.	The best result was obtained by ANN (RMSE=2.34, MAE=1.24, NSE=0.97, R2=0.97), followed by Random Forest; ML greatly increases the accuracy of WQI forecasting.	Reliance on historical data, lack of climatic elements, regional dataset limitations, and restricted physicochemical parameters. Very small dataset	262 samples (2001–2023) from the Bangladesh Water Development Board dataset; rivers: Buriganga, Turag, Balu, and Tongi Khal; parameters: DO, pH, EC, TDS, Fe, and Cl
(Han et al. 2025) [4] [Q1, Elsevier]	How to use machine learning to precisely estimate and categorize the Water Quality Index (WQI) to enhance environmental sustainability and urban waste management.	LSTM (Deep learning), Random Forest, Decision Tree, Support Vector Machine.	LSTM performed better than all other models; RF was second best; DT and SVM fared worse.	Single-municipality dataset; restricted geographic generalization; potential bias in the dataset; computational difficulty; problems with scalability	Telangana Post-Monsoon Groundwater Quality Data, a Kaggle dataset; parameters: pH, EC, TDS, and SAR; WQI divided into nine classifications (Good–Poor)
(Mohseni et al. 2024) [9] [Q1, Elsevier]	Groundwater quality must be accurately predicted using AI models because traditional water quality monitoring is expensive and ineffective.	ANN, Multiple Linear Regression, Support Vector Machine, Random Forest, XG-Boost and Stacking Ensemble Model;	All ML models obtained high prediction accuracy; the ensemble model demonstrated the best prediction accuracy, with XG-Boost outperforming the single models.	Lack of meteorological or environmental factors; small dataset size; restricted geographic coverage; computational resource requirements	54 groundwater samples were taken from 54 wards in Ujjain City, India; the sources were hand pumps, bore wells, and dug wells; the parameters were pH, turbidity, EC, TDS, alkalinity, hardness, chloride, and fluoride.
(Satish et al. 2024) [10] [Q1, Elsevier]	How can the shortcomings of traditional neural networks be overcome by a hybrid machine learning framework to increase the accuracy of Water Quality Index (WQI) prediction?	Advanced Deep Learning model (hybrid/optimized architecture) Traditional ML models: ANN, SVM, RF, and etc.,	shown stability in WQI prediction for complicated nonlinear connections.	Lack of real-time deployment, computational complexity,	Physicochemical parameters including pH, DO, BOD, COD, turbidity, temperature, and more are included in this multi-parameter water quality dataset.
(Uddin et al. 2023) [11] [Q1, Elsevier]	Different classification techniques used by the current Water Quality Index (WQI) models lead to uncertainty and inconsistency in the classification of water quality.	Total 7 WQI Models Used And Machine Learning Classifiers Used: Naïve Bayes, Support Vector Machine, and K-Nearest Neighbor. XGBoost	WQM and RMS models demonstrated more reliability, whereas XGBoost performed better than any other model. SVM and NB fared rather poorly, whereas KNN demonstrated 100% accuracy but overfitting danger.	Study restricted to a single coastal area (Cork Harbour); potential overfitting in the KNN model; reliance on the authors' earlier improved WQI approach;	catchments.ie (EPA, Ireland)
(Hridoy et al. 2025) [12] [Q1, Elsevier]	Protecting aquatic ecosystems and controlling the risks of aquatic diseases depend on accurate water quality classification and exact WQI prediction. Existing machine learning studies lack interpretability and integrated classification + regression frameworks.	Machine Learning Classification Models + Regression Models (WQI Prediction)	LightGBM and XGBoost are the best classification models. XGBoost Regressor is the best regression model. In both classification and regression tasks, ensemble gradient boosting models perform better than conventional machine learning models.	The dataset's spatiotemporal metadata is limited. Relatively modest dataset size (170 samples initially) Absence of deep learning models	Source: Mendeley Data Repository; 4300 processed records for machine learning modeling; 170 original water samples; 14 physicochemical and biological characteristics

Deep learning models, especially LSTM, consistently outperformed traditional ML methods when temporal patterns were involved. Ensemble and hybrid frameworks generally achieved the highest predictive accuracy and robustness, as summarized in Table I.

However, several important limitations remain. Many studies rely on small, region-specific datasets, restricting broader applicability and scalability. Most models operate under single-task settings, focusing only on regression or classification rather than integrating both. Environmental, climatic, and land-use factors are often insufficiently incorporated. Hybrid and ensemble models can be computationally demanding, and few studies validate their real-time deployment. Most importantly, there is still no universal, adaptive WQI framework capable of functioning across diverse water-use contexts.

Overall, the literature demonstrates clear progress in ML/DL-based WQI prediction, with ensemble and deep learning models showing superior performance. However, current approaches remain fragmented, localized, and task-specific. There is a strong need for a scalable, multitask, and adaptive AI-driven WQI architecture that integrates diverse datasets, supports multi-class classification and regression simultaneously, and generalizes across environmental contexts.

However, this literature review identified the following set

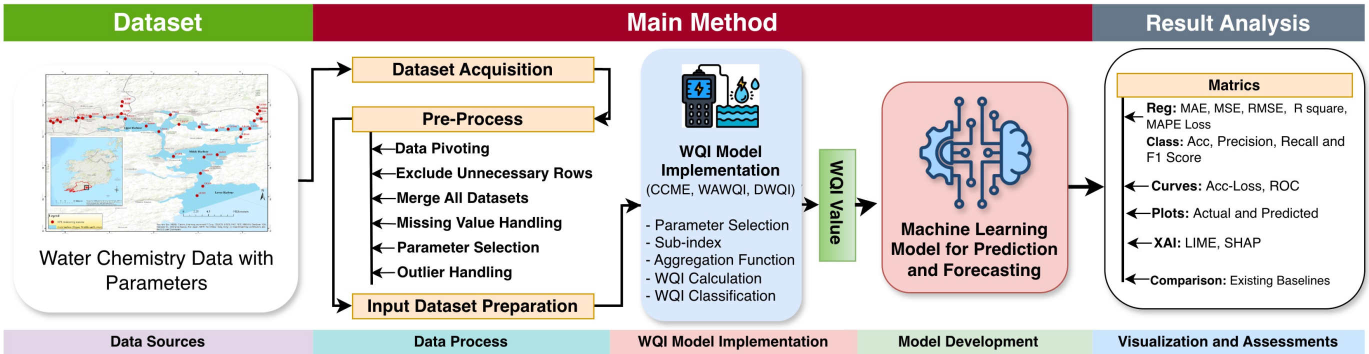


Fig. 2: Intuition and Synthesis from the Literature Review (Framework Diagram).

of research questions:

- How do machine learning models improve the prediction and classification of Water Quality Index (WQI) compared to traditional statistical methods?
- Can a generalized Water Quality Index model be developed that supports multiple water-use purposes (e.g., drinking, irrigation, ecosystem health)?
- Which machine learning approaches improve model generalization and efficiency by leveraging shared feature representations across multiple WQI models?
- How can explainable artificial intelligence (XAI) methods be integrated into machine learning-based water quality assessment?

IV. Dataset Preparation for ML based Analysis

A. Data Acquisition and Source

For this project we utilized water quality dataset available on Catchments, i.e. [16], that provides comprehensive information on Ireland's rivers, lakes, groundwater, estuaries, and coastal waters, assessed under the EU Water Framework Directive (WFD) [17].

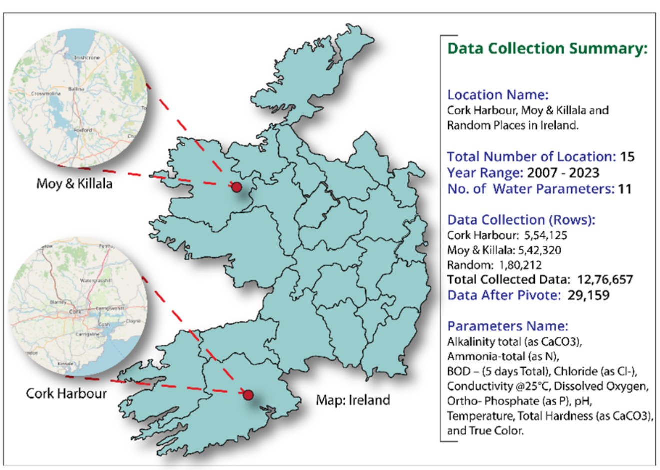
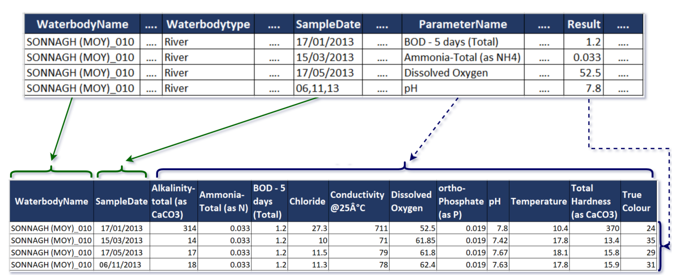


Fig. 3: Data Description and Summary

We collected initial data from the specific regions in Ireland, especially the Cork Harbour, Moy & Killala, and the other random regions, was mainly informed by the presence of complete parameters, long-term, and good-quality water records. The areas are highly varied in terms of hydro-environmental conditions, which include the coastal and riverine ecosystems, hence will provide a perfect testbed in assessing the strength of the universal non-region specific project framework.

The downloaded datasets encompasses 12,76,657 water quality data samples that spans over 15 years (from 2007 to 2023). After initial processing, 29,159 data samples are collected where each sample representing measurement of 11 selected water quality parameters for a specific day and location. The parameters were chosen according to their relevance, correlation with one another, availability, and compatibility with the WQI models referenced at [1]. This standardized data collection ensures reliability and supports transferability to other regions with minimal recalibration.

B. Data Wrangling and Pre-processing



WaterbodyName	....	Waterbodytype	....	SampleDate	....	ParameterName	....	Result	....
SONNAGH (MOY)_010	....	River	....	17/01/2013	....	BOD - 5 days (Total)	....	1.2	....
SONNAGH (MOY)_010	....	River	....	15/03/2013	....	Ammonia-Total (as NH4)	....	0.033	....
SONNAGH (MOY)_010	....	River	....	17/05/2013	....	Dissolved Oxygen	....	52.5	....
SONNAGH (MOY)_010	....	River	....	06,11,13	....	pH	....	7.8	....

WaterbodyName	SampleDate	Alkalinity-total (as CaCO3)	Ammonia-Total (as N)	BOD - 5 days (Total)	Chloride	Conductivity @25Â°C	Dissolved Oxygen	ortho-Phosphate (as P)	pH	Temperature	Total Hardness (as CaCO3)	True Colour
SONNAGH (MOY)_010	17/01/2013	314	0.033	1.2	27.3	711	52.5	0.019	7.8	10.4	370	24
SONNAGH (MOY)_010	15/03/2013	14	0.033	1.2	10	71	61.85	0.019	7.42	17.8	13.4	35
SONNAGH (MOY)_010	17/05/2013	17	0.033	1.2	11.5	79	61.8	0.019	7.67	18.1	15.8	29
SONNAGH (MOY)_010	06/11/2013	18	0.033	1.2	11.3	78	62.4	0.019	7.63	17.8	15.9	31

Fig. 4: Data Pivot Process

Data to make the data for the calculation of water quality indexing, we have to conduct two initial steps, e.g., data acquisition from the designated source and preparing the data for further analysis (as presented in Figure 3). From the official website of EPA, Ireland [16], the data are downloaded by following the given link. For each location, one CSV file is downloaded. Each data row in a CSV file represents reading for a given water quality parameter for each day between 2007 and 2023. In the first stage of data preparing, data in each CSV file is pivoted [18]. This process includes transposing the columns into rows to get the day-wise readings/measurement of the selected parameters. Then, to obtain uniformity in the parameter measurement values, the measurement units and ranges are checked against the standards [1]. This conversion of units confirms a unified and standard measurement of WQI values while applied to the WQI models. For this study, 11 water quality parameters are selected based on their relevance to the selected WQI models, availability of the parameter readings, interrelation, and accessibility [2]. Thereafter, all the CSV files are consolidated into a single CSV file that consists of 12,76,657 water quality data rows. To increase the consistency of WQI measurements by the selected models, data rows with more than three missing parameter values are removed. This refinement yields 29,159 data rows; a description is shown in Fig. 5. Finally, the missing parameters values for the remaining rows are calculated by computing median of the available value for a specific parameter. This processed dataset will then used to calculate the WQIs using the selected WQI models.

Metric	Years	Alkalinity	Ammonia	BOD	Chloride	Conductivity	Dissolved Oxygen	ortho-Phosphate	pH	Temperature	Total Hardness	True Colour
count	29159.0	27571.0	13060.0	9492.0	26491.0	24555.0	28866.0	17004.0	29148.0	28992.0	26759.0	26747.0
mean	2014.78	140.60	0.101	1.436	20.42	364.36	58.32	0.104	7.55	10.85	159.73	59.32
std	4.44	98.22	0.658	0.962	20.06	198.51	27.08	1.322	0.58	5.19	104.30	55.09
min	2007.0	0.0	0.0	0.0	0.0	33.0	0.0	-0.004	4.7	0.6	0.0	0.0
25%	2011.0	51.0	0.022	1.0	15.0	195.0	49.3	0.012	7.2	8.2	65.0	23.0
50%	2015.0	127.0	0.033	1.2	18.6	356.0	54.9	0.019	7.7	10.7	152.0	45.0
75%	2018.0	220.0	0.054	1.6	22.2	527.0	77.0	0.031	8.0	13.2	244.0	79.0
max	2023.0	442.0	40.0	16.0	1260.0	4200.0	198.0	70.0	9.8	637.0	642.0	953.0

Fig. 5: Data Description and Summary

Fig. 5 shows the descriptive statistics for main water quality metrics from 2007 to 2023. It shows that conductivity, alkalinity, hardness, and color all changed a lot. The maximum values for conductivity (4200), temperature (637), and true color (953) are all substantially higher than their median values. This suggests that there are extreme observations or possible outliers in the dataset. The count values also differ between parameters, which means that some variables may have missing (null) observations that need to be cleaned up before analysis or modeling.

Feature Name	Data Type	Non-Null Count	Null Count	Example Values
WaterbodyName	object	29159	0	ABBEYTOWN_010, Allua, ASKANAGAP STREAM_010
Years	int64	29159	0	2023, 2007, 2008
SampleDate	object	29159	0	Feb, Aug, Sep
Alkalinity-total	float64	27571	1588	314.0, 14.0, 17.0
Ammonia-Total	float64	13060	16099	0.066, 0.069, 0.068
BOD - 5 days	float64	9492	19667	1.4, 1.2, 1.1
Chloride	float64	26491	2668	27.3, 10.0, 11.5
Conductivity	float64	24555	4604	711.0, 71.0, 79.0
Dissolved Oxygen	float64	28866	293	52.5, 61.85, 61.8
ortho-Phosphate	float64	17004	12155	0.005, 0.006, 0.012
pH	float64	29148	11	7.8, 7.42, 7.67
Temperature	float64	28992	167	10.4, 17.8, 18.1
Total Hardness	float64	26759	2400	370.0, 13.4, 15.8
True Colour	float64	26747	2412	24.0, 35.0, 29.0

Fig. 6: Data Details (Feature name, Type, Missing Values, with Example Values) and Summary Before Processing.

Fig. 6 shows the structure of the dataset, the types of data it contains, and how the missing values are spread out for each water quality measure. There are no missing values in the attributes WaterbodyName, Years, and SampleDate, however there are a lot of null values in other important parameters, such as Ammonia, BOD, and ortho-Phosphate. This means that the observations are not complete. This shows that you need to preprocess your data and deal with missing values before you can do a reliable water quality study or machine learning modeling.

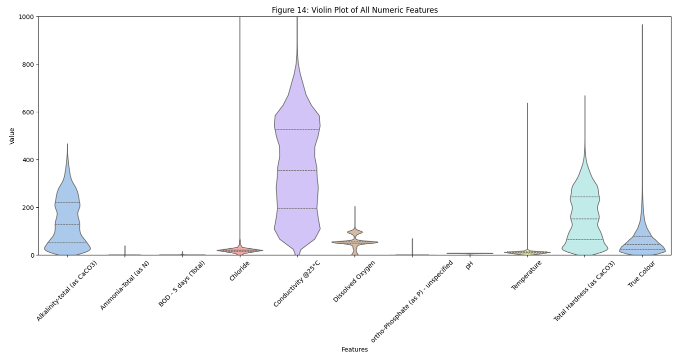


Fig. 7: Violin Plot of the Raw Dataset

Fig. 7 shows how all the numeric water quality features are spread out and how dense they are, showing how they differ from one another. Conductivity, alkalinity, hardness, and true color are examples of variables that have wider distributions and more variation. pH and ammonia, on the other hand, stay within tighter ranges. The plot also shows that some features may be skewed or have extreme values, which shows that the water quality measurements are not all the same.

This comprised dataset is highly aligned with our tentative project on multi-task machine learning based multi-context water quality index model connected with literature. This dataset contains a comprehensive set of parameters, such as pH, Dissolved Oxygen (DO), Biological Oxygen Demand (BOD), alkalinity, hardness, chloride, conductivity, temperature, nutrients etc., that required for application based water quality index calculations [1], [2]. These water quality parameters are also consistently reported in different prior studies as key inputs of water quality modelings [6], [19]. While many reviewed works that rely on small, region-specific datasets, this dataset offers a relatively large number of samples, which supports more robust training, validation, and generalization of different machine learning models. Additionally, the presence of missing values and variability across features also reflects real-world data challenges, making it suitable for testing preprocessing, imputation, and model robustness. Overall, this dataset provides a strong foundation for implementing and comparing traditional machine learning, ensemble, and deep learning approaches for scalable and generalizable WQI prediction.

Fig. 8 demonstrates that many physicochemical parameters are strongly related to each other, especially alkalinity, conductivity, and hardness. The scores connected to the WQI (IndWQI, IWQI, and DWQI) are strongly linked to a number of water quality measures, which shows that they depend on these variables. Some factors can show negative correlations, which means that the water quality measurements are related in the opposite way.

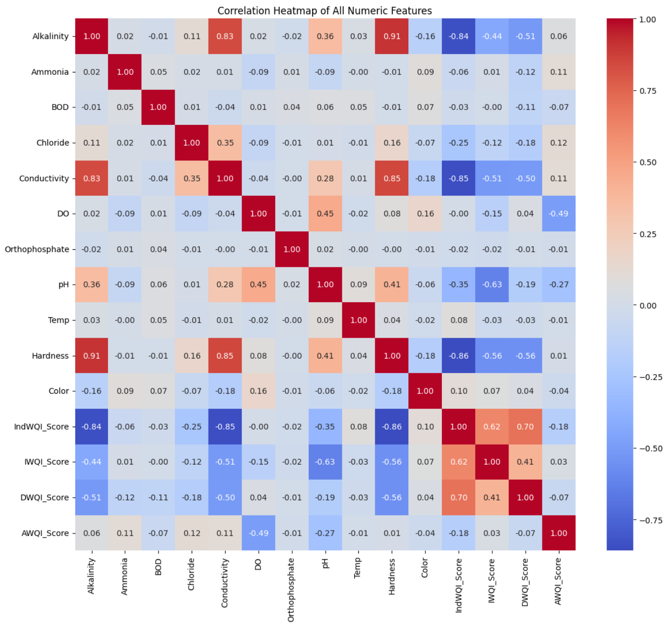


Fig. 8: Feature Correlation with WQI's of the Modified Dataset

Missing Value Handling:

After this reorganization, absent values are handled, as denoted in Figure 6. Rows with over three null entries are removed from the dataset. For the incomplete records that are left, the missing values are filled in with the median value of each corresponding parameter to ensure data consistency and robustness, utilizing the following equations.

For a data column with an odd number of sorted values, the median is determined as,

$$\text{Median} = X_{\frac{n+1}{2}} \tag{1}$$

When the column contains an even number of values, the median is calculated by averaging the two central elements:

$$\text{Median} = \frac{X_{\frac{n}{2}} + X_{\frac{n}{2}+1}}{2} \tag{2}$$

In both equations, X_i denotes the i-th element in the sorted dataset X.

Datasets with missing values generally exhibit two main patterns: Missing Completely at Random (MCAR) and Missing Not at Random (MNAR). MCAR indicates that the probability of data being absent is unrelated to both observed and unobserved factors. In contrast, MNAR indicates that the absence of data is linked to concealed variables or unobserved values. The dataset utilized in this research exhibits a missing data pattern consistent with the MCAR assumption, as no distinct trends or relationships were identified in the missing values. In these situations, median imputation serves as an effective method, as it maintains the dataset's central tendency while avoiding significant bias. Nonetheless, if the data were MNAR, more sophisticated techniques, including multiple imputation or model-driven strategies, would be necessary. Due to the unpredictable occurrence of missing values in our dataset, median imputation was used to maintain consistency and data integrity.

Data Type Conversion:

Before doing the Water Quality Index (WQI) analysis, we need to make sure that all the features in the dataset have the right data types. In the WQI dataset utilized in this research, all of the variables were already in the right format. This included both numerical physicochemical characteristics (including alkalinity, ammonia, BOD, conductivity, dissolved oxygen, pH, temperature, hardness, and color) and categorical WQI classes. So, there was no need for further preparation processes like changing date strings to datetime format, encoding categorical variables (label encoding or one-hot encoding), or dealing with mixed-type columns. This means that the WQI dataset was already well-organized and suitable for further statistical analysis and machine learning modeling.

Removing Outliers:

Outliers were identified through the z-score standardization technique, which measures how far each data point differs from the average in standard deviations. Observations exhibiting an absolute z-score ($\|z\|$) exceeding 3 were regarded as possible outliers, signaling statistically uncommon events outside the normal data distribution. Furthermore, visual examination methods, such as boxplots and scatter plots, were utilized to confirm identified anomalies and maintain data integrity. Severe variations can adversely affect the effectiveness of machine learning models, especially those that are sensitive to the scales of input features. The formula for the z-score applied in detecting anomalies is represented by the equation:

$$z = \frac{x - m(x)}{SD(x)} \tag{3}$$

Here, z "represents the standardized value", x refers to the initial feature value, $m(x)$ signifies the average of the feature, and $SD(x)$ reflects its variability. The StandardScaler() function from the scikit-learn library in Python was employed for implementation [20], adjusting each feature to achieve a mean of 0 and a standard deviation of 1. While the majority of standardized values typically fall between -3 and 3, this limit can shift based on the distribution of the data.

This sanitized dataset is prepared for computing water quality indices according to various applications. The resulting classifications are subsequently added to the dataset, forming an inclusive version that acts as the input for training, testing, and validating the suggested context-aware water quality index utilizing multi-task learning for various purposes.

The boxplots (Fig. 9 and 10) show that some water quality measures, like conductivity, chloride, color, and AWQI score, have a lot of outliers in the original dataset. This could mean that the measurements are too high or too low, or that there are problems with the data. After removing outliers, the distributions get tighter and more stable, which makes them less skewed and makes statistical analysis and machine learning modeling more reliable.

Feature Engineering

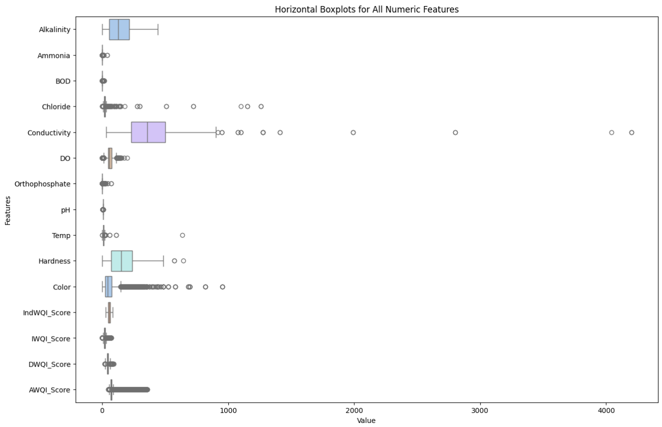


Fig. 9: Dataset before Removing Outliers.

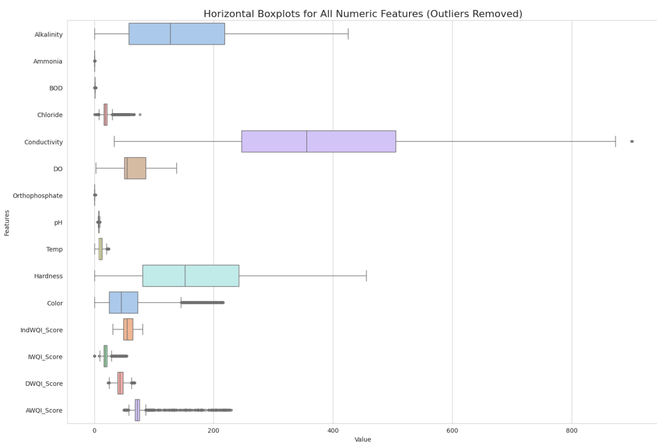


Fig. 10: Dataset after Removing Outliers.

Derived Features for Water Quality Dataset: BOD to DO Ratio

The BOD to DO ratio shows how much oxygen is needed in water compared to how much dissolved oxygen is available. High values mean that the water is stressed for oxygen, which can mean that the water quality is not good.

$$\text{BOD\_DO\_Ratio} = \frac{\text{BOD}}{\text{DO} + \epsilon} \tag{4}$$

where, BOD - Biochemical Oxygen Demand (mg/L), DO - Dissolved Oxygen (mg/L), and ϵ - A small number is used to prevent the division by zero problem.

Rationale for ML Models:

- A single feature is created by combining two water quality indicators.
- To improve the sensitivity of the model and make the WQI prediction more accurate.

Hardness to Alkalinity Ratio

This ratio tells how much water hardness affects alkalinity. It shows how likely it is for water to scale and how well it can buffer.

$$\text{Hardness\_Alkalinity\_Ratio} = \frac{\text{Hardness}}{\text{Alkalinity} + \epsilon} \tag{5}$$

where, Hardness - The total hardness of water (mg/L as $CaCO_3$), Alkalinity - Alkalinity of water (mg/L as $CaCO_3$), and ϵ - a small number is used to prevent the division by zero problem.

Rationale for ML Models:

- Two chemical features interact to each other.
- Helpful for finding patterns of water scaling or buffering that change the quality of the water as a whole.

C. Water Quality Index Calculation

WQIs are mathematical models created to convert various intricate water quality parameter measurements into a single, dimensionless value that clearly conveys the overall condition of water quality. This straightforward depiction facilitates interpretation and action for policymakers, researchers, and the general public [1], [2]. Utilizing specialized water quality indexing based on use is essential for guaranteeing the safe and effective use of water resources by customizing evaluations to meet the distinct requirements of various applications.

This study employs four specific WQI models, including the Drinking Water Quality Index (DWQI) [21], the Irrigation Water Quality Index (IWQI) [22], the Industrial Water Quality Index (IndWQI) [23], and the Aquatic Water Quality Index (AQWI) [24] [25]. These models were selected because of their widespread application in specific research and practical uses pertaining to water quality evaluation [1]. Typically, WQI models adhere to four fundamental steps executed in order to generate the WQI, which include: (a) choosing water quality parameters, (b) generating the sub-indices for those parameters, (c) assigning weights to the parameters, and ultimately, (d) using an aggregation function to compute the final index [1].

	Feature Name	Data Type	Non-Null Count	Null Count	Unique Values	Example Values
0	WaterbodyName	object	25633	0	158	ABBEYTOWN_010, Allua, ASKANAGAP STREAM_010
1	Years	int64	25633	0	17	2023, 2007, 2008
2	SampleDate	object	25633	0	12	Feb, Aug, Sep
3	Alkalinity	float64	25633	0	501	0.74, 0.04, 0.04
4	Ammonia	float64	25633	0	238	0.02, 0.05, 0.03
5	BOD	float64	25633	0	170	0.41, 0.48, 0.38
6	Chloride	float64	25633	0	808	0.36, 0.15, 0.15
7	Conductivity	float64	25633	0	783	0.78, 0.05, 0.05
8	DO	float64	25633	0	1711	0.37, 0.44, 0.44
9	Orthophosphate	float64	25633	0	155	0.01, 0.00, 0.00
10	pH	float64	25633	0	325	0.60, 0.57, 0.55
11	Temp	float64	25633	0	222	0.41, 0.73, 0.72
12	Hardness	float64	25633	0	904	0.81, 0.03, 0.03
13	Color	float64	25633	0	218	0.11, 0.13, 0.14
14	IndWQI	object	25633	0	3	Poor, Good, Moderate
15	IndWQI_Score	float64	25633	0	12906	0.23, 0.94, 0.94
16	IWQI	object	25633	0	2	Very Bad, Bad
17	IWQI_Score	float64	25633	0	1572	0.32, 0.95, 0.96
18	DWQI	object	25633	0	3	Poor, Medium, Very Poor
19	DWQI_Score	float64	25633	0	10334	0.29, 0.67, 0.67
20	AWQI	object	25633	0	4	Good, Regular, Excellent
21	AWQI_Score	float64	25633	0	1009	0.12, 0.04, 0.16

Fig. 11: Data Summary After WQI Calculation.

The correlation heatmap (Fig. 11) shows that a lot of physicochemical factors are closely related to each other, especially alkalinity, conductivity, and hardness. The scores that go with the WQI (IndWQI, IWQI, and DWQI) are very closely related to a number of indicators of water quality, which suggests that they depend on these variables. Some factors can show negative correlations, which means that the measures of water quality are related in the opposite way.

Data Normalization/Scaling

In Water Quality Index (WQI) modeling, different parameters have different values and ranges. So, you can't directly assemble the raw values or utilize them for WQI aggregation.

Min–max normalization is a common approach to address this issue:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{6}$$

where x- the original value of the parameter, $x_{\min}$ and $x_{\max}$- the minimum and maximum values of the feature, and x'- the normalized value scaled between 0 and 1.

Why Min–Max Normalization is Used in WQI:

In this analysis, Min-Max normalization was applied to scale all numeric features to a consistent range of 0 to 1. This technique is particularly suitable when preparing data for deep learning models, which are sensitive to the scale of input features; large differences in feature magnitudes can slow convergence or lead to unstable training. This method has the following benefits:

- Compares Different Parameters
- To prevent dominance of the Feature
- Ensures Data Distribution
- Helpful for combining many WQI parameters
- Makes multi-task model convergence better
- Supports interpretability techniques like SHAP.

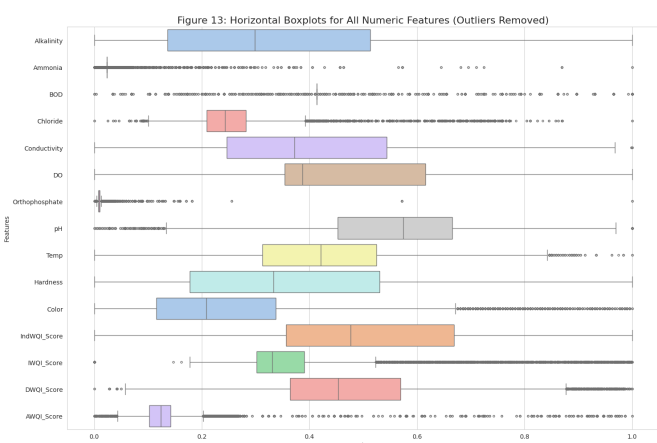
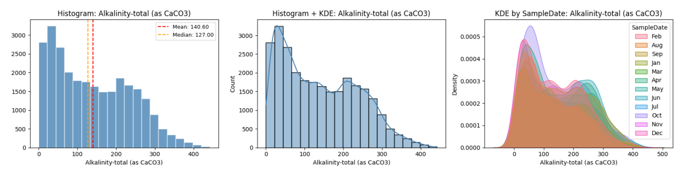


Fig. 12: Feature Representation After Scaling/Normalization

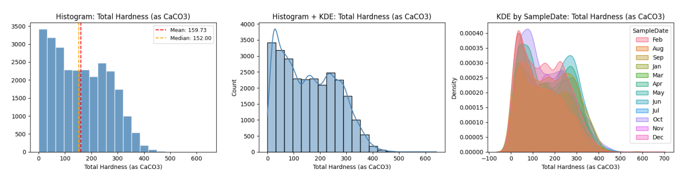
Before scaling, the boxplots (Fig. **??**) demonstrate that the individual water quality metrics have significantly diverse value ranges. Conductivity, hardness, and alkalinity are the most important variables on the scale, which makes it impossible to compare them. All features are put into a comparable range after normalization (Fig. 12), which makes the distribution more even. This lets each parameter play a fair role in machine learning and statistical analysis.

Min-Max scaling keeps the original distribution and relationships between values, which lets the model use the whole numeric range. This is different from RobustScaler, which uses the interquartile range to reduce the effect of outliers. Min-Max normalization is acceptable here because our dataset has previously been cleaned up to remove or deal with outliers. This way, each feature will help the learning process in a fair way. This strategy makes gradient-based optimization better, speeds up convergence during training, and helps the deep learning framework learn complex non-linear patterns well. If standard scaling methods made for datasets with a lot of outliers were employed, this might not be possible.

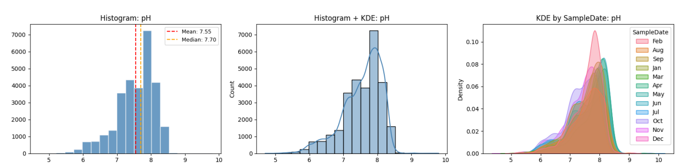
subcaption



(a) Dashboard for Alkalinity



(b) Dashboard for Hardness



(c) Dashboard for pH

Fig. 13: Feature Distribution for Top 3 Features

Fig. 13 shows the distribution and density of important water quality measures, such as alkalinity, hardness, and pH. Alkalinity and hardness have larger and right-skewed distributions, which means they are more variable. On the other hand,pH readings are mostly around neutral levels (around 7–8). The monthly KDE curves also show that there are small seasonal changes between sampling months.

V. Tentative Method Planned for the Project

A simple or hybrid machine learning model will be used to predict application and user based WQI and classify water quality for different purposes, as depicted in Figure 14.

This framework will provide an adaptive WQI framework that will provide the following features:

- Dynamic parameter weighting based on use-case (e.g., drinking, irrigation).

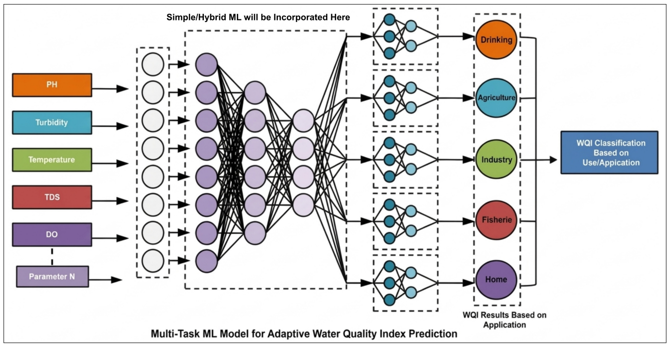


Fig. 14: Tentative Multi-task ML Project Framework

- Learning-based calibration of thresholds using historical and real-time data.
- Explainable AI features (e.g., LIME, SHAP) to provide transparency in decision-making.
- Water Hazard Analysis of an area with future WQI forecasting.

VI. Synthesis and Conclusion

Water Quality Index (WQI) models have developed to condense complex environmental information into a single representative score. Typically, WQI models adhere to four fundamental steps executed in order to generate the WQI, which include: (a) choosing water quality parameters, (b) generating the sub-indices for those parameters, (c) assigning weights to the parameters, and ultimately, (d) using an aggregation function to compute the final index [1]. An intuitive summary of the literature synthesis is depicted in Figure 2.

Machine learning is data-driven approach that can capture complex, nonlinear relationships among water quality parameters and predict accurate index without relying on fixed mathematical assumptions. Table I illustrated an exclusive summary on existing studies. Different machine learning and deep learning models are incorporated to predict water quality indices, refereed to column *Methods* in Table I have prediction accuracy ranged from 84% to 99% in different dataset for a single water quality index. However, these models are mostly used for estimating single purpose water quality index. On the other hand, most existing research overlook interpretablity, where the explainable artificial intelligence (XAI) helps not only to visualize the parameters responsible for a prediction, but also assist verify which parameters suppose to contribute for an accurate prediction [26]. In real life environment, classification of water quality index play a vital role. In the previous work most of the study solely focus on either binary classification or merging different classes, emphasizing less on real classification of the water quality index models.

In machine learning based evaluation, dataset play a vital role. In order to calculate WQIs, parameters selection is crucial step as the different indexing models have different set of input parameters. In the existing exploration we found a large collection of water quality datasets. But, only few datasets have the required parameters to calculate different water quality indices. Amon all the datasets, we found Ireland Catchments dataset have the all required parameters with a vast collection data items. It provides comprehensive information on Ireland's rivers, lakes, groundwater, estuaries, and coastal waters, assessed under the EU Water Framework Directive (WFD). This standardized data collection ensures reliability and supports transferability to other regions with minimal recalibration.

Our project will focus on addressing a crucial limitation of the need of an universal water quality index, incorporating machine learning, where most current WQI models are developed for specific regions, such as Canada or Ireland, or tailored to particular water sources like rivers or wetlands. These models typically generate a single index using predefined parameter weights and fixed threshold values. As a result, they lack the flexibility required for multi-use environments where the same water body may serve drinking, irrigation, fisheries, and other purposes simultaneously. This limitation highlights an important research gap: the need for a more universal and adaptive AI-driven WQI framework capable of supporting large-scale application across diverse environmental contexts while aligning with modern sustainability requirements.

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